

Model Question Paper

Network Security

Time : 3 Hour

Max.Marks : 75

PART A

I. Answer all the following questions (9 x 1 = 9 Marks)

1	What is the inter processor distance in LAN?	M 1.02	U
2	Which layer is responsible for moving frames from one node to the next?	M 1.04	R
3	When a switch receives traffic from a port with a MAC source address, it binds the port number and that MAC address. The ----- attack in the Data Link Layer exploits this ability of the switches.	M 2.02	A
4	Give the names of the three parameters by which a security association is uniquely identified.	M 2.11	R
5	Identify the type of attack that involves persuading a user and an access point to believe that they are talking to each other when in fact the communication is going through an intermediate attacking device.	M 2.05	U
6	What is server write key and client write key in a TLS session state?	M 3.03	R
7	Write the names of any two attacks that can be made on packet filtering firewalls.	M 4.06	R
8	Name the two schemes developed for E-mail security.	M 3.08	R
9	Which are the elements of a Network Access Control system?	M 4.01	R

PART B

II Answer any 8 questions from the following .Each questions carry 3 marks.

(8 x 3 = 24 Marks)

1	Why do we need Network security?	M 1.06	U
2	What do you mean by access control in a network?	M 1.07	R
3	What are the measures to ensure security in Wireless LAN?	M 2.03	U
4	What are the applications of IPSec? Where does IPsec reside in a protocol stack?	M 2.09	R
5	Identify any three security threats in mobile devices.	M 2.07	U
6	What are the four principal services provided by S/MIME?	M 3.11	R
7	Differentiate between an SSL connection and an SSL session? List and briefly define the parameters that define an SSL session state.	M 3.02	U
8	Briefly describe any three security issues in internet banking.	M4.09	U
9	Briefly define four EAP authentication methods and its working.	M 4.03	U
10	Differentiate Application gateways and circuit level gateways.	M4.07	U

PART C

Answer ALL questions. Each carry 7 marks.

(6 x 7 = 42 Marks)

1.	How are computer networks classified based on the topology? Explain each with suitable figure. OR	M 1.01	U
2.	What is meant by a Network Reference model? Explain ISO/OSI reference model.	M 1.03	R
3.	Explain the basic model for Network Security. OR	M 1.08	U
4.	Explain the OSI security architecture in detail.	M 1.02	U
5.	What is Wireless Access Protocol (WAP)? Explain. OR	M 2.06	U
6.	Which are the different IPsec Communication Modes? Also explain the IP security policy.	M 2.10	R

7.	Explain the SSH Transport Layer Protocol in detail with suitable diagrams. O R	M3.06	U
8.	Explain the operation of PGP Scheme.	M3.10	U
9.	Explain in detail the TLS architecture and its protocol stack. OR	M3.03	U
10.	Explain the operation of DNS and DNSSEC.	M3.12	U
11. .	How does IEEE 802.1X Port-Based Network Access Control provide access control functions for LANs? O R	M4.04	U
12. .	What is a DMZ network and what types of systems would you expect to find on such networks?	M4.08	A